

```
package com.jdbc;
```

```
import java.sql.*;
```

```
import java.util.Scanner;
```

```
public class BusReservationSystem {
```

```
    static Scanner sc = new Scanner(System.in);
```

```
    // ----- VALIDATIONS -----
```

```
    public static boolean validName(String name) {
```

```
        return name.matches("[A-Za-z ]+");
```

```
    }
```

```
    public static boolean validMobile(String mobile) {
```

```
        return mobile.matches("[0-9]{10}");
```

```
    }
```

```
    // ----- MAIN -----
```

```
    public static void main(String[] args) {
```

```
        try {
```

```
            Class.forName("com.mysql.cj.jdbc.Driver");
```

```
            Connection con = DriverManager.getConnection(
```

```
"jdbc:mysql://localhost:3306/jdbc_project",  
"root",  
"Shalu@1302");
```

```
System.out.println("Database connected successfully!");
```

```
createTables(con);
```

```
System.out.println("Tables created successfully!");
```

```
int choice;
```

```
do {
```

```
    System.out.println("\n--- BUS RESERVATION SYSTEM ---");
```

```
    System.out.println("1. Search Buses");
```

```
    System.out.println("2. Add Passenger");
```

```
    System.out.println("3. Add Bus");
```

```
    System.out.println("4. Book Ticket");
```

```
    System.out.println("5. Display Bookings");
```

```
    System.out.println("6. Cancel Ticket");
```

```
    System.out.println("7. Custom Query");
```

```
    System.out.println("0. Exit");
```

```
    System.out.print("Enter choice: ");
```

```
    choice = sc.nextInt();
```

```
    sc.nextLine();
```

```
switch (choice) {
```

```
    case 1:
```

```
        searchBuses(con);
```

```
        break;
```

```
    case 2:
```

```
        addPassenger(con);
```

```
        break;
```

```
    case 3:
```

```
        addBus(con);
```

```
        break;
```

```
    case 4:
```

```
        bookTicket(con);
```

```
        break;
```

```
    case 5:
```

```
        displayBookings(con);
```

```
        break;
```

```
    case 6:
```

```
        cancelTicket(con);
```

```
        break;
```

```
    case 7:
```

```
        customQuery(con);
```

break;

case 0:

System.**out**.println("Exiting...");

break;

default:

System.**out**.println("Invalid choice!");

}

} **while** (choice != 0);

con.close();

} **catch** (Exception e) {

e.printStackTrace();

}

}

// ----- CREATE TABLES -----

static void createTables(Connection con) **throws** Exception {

Statement st = con.createStatement();

st.executeUpdate("""

```
CREATE TABLE IF NOT EXISTS buses(  
    bus_id INT PRIMARY KEY AUTO_INCREMENT,  
    bus_name VARCHAR(50) NOT NULL,  
    source VARCHAR(50) NOT NULL,  
    destination VARCHAR(50) NOT NULL,  
    total_seats INT NOT NULL CHECK(total_seats > 0),  
    available_seats INT NOT NULL  
)  
""");
```

```
st.executeUpdate("""
```

```
CREATE TABLE IF NOT EXISTS passengers(  
    passenger_id INT PRIMARY KEY AUTO_INCREMENT,  
    name VARCHAR(50) NOT NULL,  
    email VARCHAR(100) UNIQUE NOT NULL,  
    mobile VARCHAR(10) NOT NULL,  
    age INT NOT NULL CHECK(age > 0 AND age < 100)  
)  
""");
```

```
st.executeUpdate("""
```

```
CREATE TABLE IF NOT EXISTS bookings(  
    booking_id INT PRIMARY KEY AUTO_INCREMENT,  
    passenger_id INT,  
    bus_id INT,  
    seat_no INT NOT NULL,  
    FOREIGN KEY(passenger_id)  
    REFERENCES passengers(passenger_id),
```

```
        FOREIGN KEY(bus_id)
        REFERENCES buses(bus_id)
    )
    """);
```

```
    st.close();
}
```

```
// ----- SEARCH BUSES -----
```

```
static void searchBuses(Connection con) {
```

```
    try {
```

```
        System.out.print("Enter Source: ");
```

```
        String source = sc.nextLine();
```

```
        System.out.print("Enter Destination: ");
```

```
        String destination = sc.nextLine();
```

```
        PreparedStatement ps = con.prepareStatement(
```

```
            "SELECT bus_id, bus_name, source, destination, " +
```

```
            "total_seats, available_seats " +
```

```
            "FROM buses WHERE source=? AND destination=?");
```

```
        ps.setString(1, source);
```

```
        ps.setString(2, destination);
```

```
ResultSet rs = ps.executeQuery();
```

```
System.out.println("\n--- AVAILABLE BUSES ---");
```

```
boolean found = false;
```

```
while (rs.next()) {
```

```
    found = true;
```

```
    System.out.println(
```

```
        "Bus ID: " + rs.getInt("bus_id")
```

```
        + " | Bus: " + rs.getString("bus_name")
```

```
        + " | Route: " + rs.getString("source")
```

```
        + " -> " + rs.getString("destination")
```

```
        + " | Total Seats: " + rs.getInt("total_seats")
```

```
        + " | Available Seats: "
```

```
        + rs.getInt("available_seats"));
```

```
    }
```

```
if (!found) {
```

```
    System.out.println("No buses found for this route!");
```

```
}
```

```
rs.close();
```

```
ps.close();
```

```
} catch (Exception e) {
```

```
        System.out.println("Search Error: " + e.getMessage());
    }
}
```

```
// ----- ADD PASSENGER -----
```

```
static void addPassenger(Connection con) {
```

```
    try {
```

```
        System.out.print("Enter Name: ");
```

```
        String name = sc.nextLine();
```

```
        System.out.print("Enter Email: ");
```

```
        String email = sc.nextLine();
```

```
        System.out.print("Enter Mobile (10 digits): ");
```

```
        String mobile = sc.nextLine();
```

```
        System.out.print("Enter Age: ");
```

```
        int age = sc.nextInt();
```

```
        sc.nextLine();
```

```
        if (!validName(name)) {
```

```
            System.out.println("Invalid Name!");
```

```
            return;
```

```
        }
```

```
if (!email.contains("@")) {  
    System.out.println("Invalid Email!");  
    return;  
}
```

```
if (!validMobile(mobile)) {  
    System.out.println("Invalid Mobile!");  
    return;  
}
```

```
if (age <= 0 || age >= 100) {  
    System.out.println("Invalid Age!");  
    return;  
}
```

```
PreparedStatement ps = con.prepareStatement(  
    "INSERT INTO passengers(name,email,mobile,age) " +  
    "VALUES(?,?,?,?)");
```

```
ps.setString(1, name);  
ps.setString(2, email);  
ps.setString(3, mobile);  
ps.setInt(4, age);
```

```
ps.executeUpdate();
```

```
ps.close();
```

```
        System.out.println("Passenger added successfully!");

    } catch (Exception e) {

        System.out.println("Error: " + e.getMessage());
    }
}

// ----- ADD BUS -----

static void addBus(Connection con) {

    try {

        System.out.print("Enter Bus Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Source: ");
        String source = sc.nextLine();

        System.out.print("Enter Destination: ");
        String destination = sc.nextLine();

        System.out.print("Enter Total Seats: ");
        int seats = sc.nextInt();
        sc.nextLine();
```

```
if (seats <= 0) {  
    System.out.println("Invalid number of seats!");  
    return;  
}
```

```
PreparedStatement ps = con.prepareStatement(  
    "INSERT INTO buses(bus_name,source,destination," +  
    "total_seats,available_seats) VALUES(?,?,?,?,?)");
```

```
ps.setString(1, name);  
ps.setString(2, source);  
ps.setString(3, destination);  
ps.setInt(4, seats);  
ps.setInt(5, seats);
```

```
ps.executeUpdate();
```

```
ps.close();
```

```
System.out.println("Bus added successfully!");
```

```
} catch (Exception e) {
```

```
    System.out.println("Error: " + e.getMessage());
```

```
}
```

```
}
```

```
// ----- BOOK TICKET -----
```

```
static void bookTicket(Connection con) {

    try {

        System.out.print("Enter Passenger ID: ");

        int pid = sc.nextInt();

        System.out.print("Enter Bus ID: ");

        int bid = sc.nextInt();

        System.out.print("Enter Seat No: ");

        int seat = sc.nextInt();

        sc.nextLine();

        con.setAutoCommit(false);

        // Check Passenger

        PreparedStatement passengerCheck = con.prepareStatement(
            "SELECT passenger_id FROM passengers WHERE passenger_id=?");

        passengerCheck.setInt(1, pid);

        ResultSet passengerResult = passengerCheck.executeQuery();

        if (!passengerResult.next()) {
```

```
System.out.println("Passenger ID not found!");
```

```
con.rollback();
```

```
con.setAutoCommit(true);
```

```
passengerResult.close();
```

```
passengerCheck.close();
```

```
return;
```

```
}
```

```
passengerResult.close();
```

```
passengerCheck.close();
```

```
// Check Bus
```

```
PreparedStatement busCheck = con.prepareStatement(
```

```
    "SELECT total_seats, available_seats " +
```

```
    "FROM buses WHERE bus_id=?");
```

```
busCheck.setInt(1, bid);
```

```
ResultSet busResult = busCheck.executeQuery();
```

```
if (!busResult.next()) {
```

```
    System.out.println("Bus ID not found!");
```

```
con.rollback();

con.setAutoCommit(true);


busResult.close();

busCheck.close();


return;
}


int totalSeats = busResult.getInt("total_seats");
int availableSeats = busResult.getInt("available_seats");


busResult.close();

busCheck.close();


// Check Seat Number


if (seat < 1 || seat > totalSeats) {


    System.out.println(
        "Invalid Seat No! Enter between 1 and "
        + totalSeats);


con.rollback();

con.setAutoCommit(true);


return;
}
```

```
// Check Available Seats
```

```
if (availableSeats <= 0) {
```

```
    System.out.println("No seats available!");
```

```
    con.rollback();
```

```
    con.setAutoCommit(true);
```

```
    return;
```

```
}
```

```
// Check Already Booked Seat
```

```
PreparedStatement seatCheck = con.prepareStatement(
```

```
    "SELECT booking_id FROM bookings " +
```

```
    "WHERE bus_id=? AND seat_no=?");
```

```
seatCheck.setInt(1, bid);
```

```
seatCheck.setInt(2, seat);
```

```
ResultSet seatResult = seatCheck.executeQuery();
```

```
if (seatResult.next()) {
```

```
    System.out.println("Seat already booked!");
```

```
con.rollback();

con.setAutoCommit(true);


seatResult.close();

seatCheck.close();


return;
}


seatResult.close();

seatCheck.close();


// Insert Booking

PreparedStatement bookingStatement = con.prepareStatement(
    "INSERT INTO bookings(passenger_id,bus_id,seat_no) " +
    "VALUES(?,?,?)");


bookingStatement.setInt(1, pid);
bookingStatement.setInt(2, bid);
bookingStatement.setInt(3, seat);


bookingStatement.executeUpdate();


bookingStatement.close();


// Reduce Available Seats
```

```
PreparedStatement seatUpdate = con.prepareStatement(
    "UPDATE buses " +
    "SET available_seats = available_seats - 1 " +
    "WHERE bus_id=?");

seatUpdate.setInt(1, bid);

seatUpdate.executeUpdate();

seatUpdate.close();

// Commit Transaction

con.commit();

con.setAutoCommit(true);

System.out.println("Ticket booked successfully!");

} catch (Exception e) {

    try {

        con.rollback();

        con.setAutoCommit(true);

    } catch (Exception ex) {
```

```

        ex.printStackTrace();
    }

    System.out.println("Booking Error: " + e.getMessage());
}
}

```

```
// ----- DISPLAY BOOKINGS -----
```

```
static void displayBookings(Connection con) {
```

```
    try {
```

```
        Statement st = con.createStatement();
```

```
        ResultSet rs = st.executeQuery("""
```

```
            SELECT b.booking_id,
```

```
                p.name,
```

```
                bu.bus_name,
```

```
                bu.source,
```

```
                bu.destination,
```

```
                b.seat_no
```

```
        FROM bookings b
```

```
        JOIN passengers p
```

```
        ON b.passenger_id=p.passenger_id
```

```
        JOIN buses bu
```

```
        ON b.bus_id=bu.bus_id
```

```
        """);
```

```
System.out.println("\n--- ALL BOOKINGS ---");
```

```
boolean found = false;
```

```
while (rs.next()) {
```

```
    found = true;
```

```
    System.out.println(
```

```
        "Booking ID: " + rs.getInt("booking_id")
```

```
        + " | Passenger: " + rs.getString("name")
```

```
        + " | Bus: " + rs.getString("bus_name")
```

```
        + " | Route: " + rs.getString("source")
```

```
        + " -> " + rs.getString("destination")
```

```
        + " | Seat: " + rs.getInt("seat_no"));
```

```
    }
```

```
if (!found) {
```

```
    System.out.println("No bookings found!");
```

```
}
```

```
rs.close();
```

```
st.close();
```

```
} catch (Exception e) {
```

```
    System.out.println("Error: " + e.getMessage());
```

```
}  
}
```

```
// ----- CANCEL TICKET -----
```

```
static void cancelTicket(Connection con) {
```

```
    try {
```

```
        System.out.print("Enter Booking ID to cancel: ");
```

```
        int id = sc.nextInt();
```

```
        sc.nextLine();
```

```
        con.setAutoCommit(false);
```

```
        // Find Bus ID
```

```
        PreparedStatement findStatement = con.prepareStatement(
```

```
            "SELECT bus_id FROM bookings WHERE booking_id=?");
```

```
        findStatement.setInt(1, id);
```

```
        ResultSet result = findStatement.executeQuery();
```

```
        if (!result.next()) {
```

```
            System.out.println("Booking ID not found!");
```

```
con.rollback();

con.setAutoCommit(true);


result.close();

findStatement.close();


return;
}


int busId = result.getInt("bus_id");


result.close();

findStatement.close();


// Delete Booking


PreparedStatement deleteStatement = con.prepareStatement(
    "DELETE FROM bookings WHERE booking_id=?");


deleteStatement.setInt(1, id);


deleteStatement.executeUpdate();


deleteStatement.close();


// Increase Available Seats


PreparedStatement seatUpdate = con.prepareStatement(
```

```
"UPDATE buses " +
```

```
"SET available_seats = available_seats + 1 " +
```

```
"WHERE bus_id=?");
```

```
seatUpdate.setInt(1, busId);
```

```
seatUpdate.executeUpdate();
```

```
seatUpdate.close();
```

```
con.commit();
```

```
con.setAutoCommit(true);
```

```
System.out.println("Ticket cancelled successfully!");
```

```
} catch (Exception e) {
```

```
    try{
```

```
        con.rollback();
```

```
        con.setAutoCommit(true);
```

```
    } catch (Exception ex) {
```

```
        ex.printStackTrace();
```

```
    }
```

```
        System.out.println("Cancel Error: " + e.getMessage());
    }
}
```

```
// ----- CUSTOM QUERY -----
```

```
static void customQuery(Connection con) {
```

```
    try {
```

```
        Statement st = con.createStatement();
```

```
        ResultSet rs = st.executeQuery("""
```

```
            SELECT bu.bus_name,
```

```
                COUNT(b.booking_id) AS total
```

```
            FROM buses bu
```

```
            LEFT JOIN bookings b
```

```
            ON bu.bus_id=b.bus_id
```

```
            GROUP BY bu.bus_id, bu.bus_name
```

```
            """);
```

```
        System.out.println("\n--- BUS BOOKING COUNT ---");
```

```
        while (rs.next()) {
```

```
            System.out.println(
```

```
                "Bus: " + rs.getString("bus_name")
```

```
                + " | Total Bookings: "
```

```
        + rs.getInt("total"));
    }

    rs.close();
    st.close();

} catch (Exception e) {

    System.out.println("Error: " + e.getMessage());
}
}
}
```